www.csvtuonline.com

Printed Pages – 8

325356 (24)

BE (3rd Semester)

Examination, April-May, 2014

Branch : Electrical & Electronics Engineering

ELECTRICAL MACHINES - I (NEW)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt part (a) and any two parts from (b), (c) and (d) of each question.

Unit-I

Q. 1. (a) Why a transformer can not operate on d.c supply ? Explain 2

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(b) Derive E.M.F. equation of a transformer. A single phase transformer has 200 primary and 500 secondary turns. The net cross sectional area of the core is 50 cm^2. If the primary winding be connected to a 50Hz supply at 500 V, calculate : 3+4

(i) The peak value of flux density in the core

(ii) The voltage induced in the secondary winding

(c) What are the different losses occur in a transformer on load ? How can these losses be determined ? Explain with diagram. 7

www.csvtuonline.com

www.csvtuonline.com

(d) Draw the phasor diagram of a loaded single phase transformer at lagging p.f. and derive from this equivalent circuit of the transformer. 7

Unit-II

Q. 2. (a) State the necessary conditions for satisfactory operation of two transformers in parallel. 2

(b) Explain the working principle and construction of an auto transformer. 7

(c) A transformer has its maximum efficiency of 0.98 at 15 KVA at unity p.f. During a day it is loaded as follows : 7

12 hours : 2 kW at 0.8 pf

6 hours : 12 kW at 0.6 pf

6 hours : 18 kW at 0.9 pf

Find the all day efficiency.

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(d) Two single phase transformers A and B rated at 500 KVA ech are operated in parallel percentage impedances for A and B are (1 + j6) and (1.2 + j4.8) respectively. Compute the load shared by each when the total load is 1000 KVA at 0.8 p.f. lagging. 7

Unit-III

Q. 3. (a) State the different vector groups. 2

(b) Draw the diagram of scott connection of transformers. Explain its working. 7

(c) A 3 phase step down transformer is connected to 6.6 kV mains and takes 10 A. Calculate the secondary line voltage, line current and out put for the

www.csvtuonline.com

www.csvtuonline.com

(i) Δ / Y and Y / Δ connections

The ratio of turns per phase is 12. Neglect losses. 7

(d) A 3-phase transformer, ratio 33/6.6 kv, Δ / Y , 2 MVA has a primary resistance of 8 Ω per phase and a secondary resistance of 0.08 ohm per phase. The percentage impedance is 7%. Calculate the secondary voltage with rated primary voltage and the regulation for full load 0.8 p.f. lagging. 7

Unit-IV

Q. 4. (a) What is BLV concept ? 2

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(b) State the classification of d.c. generators and explain with diagram. 7

(c) What is armature reaction ? What are its effects ? How will you neutralise these effects ? 7

(d) The magnetization characteristic for a 4 pole, 110 V, 1000 r.p.m. shunt generator is as follows : 7

Field current (A)	0	0.5	1	1.5	2	2.5	3
O.C. voltage (V)	5	50	85	102	112	116	120

Armature is lap connected with 144 conductors. Field resistance is 45 ohms. Determine :

www.csvtuonline.com

www.csvtuonline.com

(i) Voltage the machine build up of no load

(ii) The critical resistance

(iii) The speed at which the machine just fails to excite

(iv) Residual flux per pole

Unit-V

Q. 5. (a) Why should a d.c. series motor not be run without load ? 2

(b) What is the need of starter ? Explain three point starter with all protecting devices. 7

(c) Derive and draw the characteristics of DC shunt and series motor with the help of basic equation. 7

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(d) A 250 volt d.c. shunt motor has armature resistance of 0.25 ohm, on load it takes an armature current of 50 A and runs at 750 r.p.m. If the flux of motor is reduced by 10% without changing the load torque, find the new speed of the motor. 7

www.csvtuonline.com

www.csvtuonline.com